# Dynamic Evolution: Continuous Change Monitoring in Time-Series Remote Sensing Images

Jialu Li, Lingyu Sun, Chen Wu, *Member, IEEE*, Bo Du, *Senior Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*—The Earth's surface experiences continuous changes across multiple spatial and temporal scales, and time-series remote sensing images (TSIs) provide continuous observation data for monitoring and analyzing these changes. However, most existing time-series change detection (TSCD) methods based on deep learning primarily focus on binary change classification, lacking the ability to capture the temporal evolution of changes. This limitation hinders a comprehensive understanding of complex, time-dependent change processes. To address this issue, we define continuous change monitoring as the dynamic tracking of changes across multiple time points, which enables not only the detection of change occurrence but also the precise determination of when these changes occur in TSIs. To bridge the methodological gap in deep learning-based continuous change monitoring, we propose a novel continuous change monitoring network (CCM-Net) designed to capture temporal change dynamics in TSIs. CCM-Net employs a U-Net-based convolutional neural network to extract spatial features and generate difference features that emphasize temporal variations. Furthermore, a spatial-spatiotemporal LSTM (SSL) module is incorporated to associate similar change patterns across different spatial locations and time steps. A monitoring inference moment (MIM) module is also introduced to infer the moment of changes, integrating continuous change monitoring and change moment identification into a unified framework. The proposed approach is evaluated on two global-scale datasets, DynamicEarthNet and SpaceNet7. Experimental results demonstrate that CCM-Net achieving $F1$-score improvements of 7.31% and 2.84% over the best competing method on DynamicEarthNet and SpaceNet7, respectively, indicating its effectiveness in tracking temporal evolution of changes in TSIs. Our source code will be made publicly available at: https://github.com/lijialu144/CCM-Net**

***Index Terms*—Change moment identification, continuous change monitoring, deep learning, time-series remote sensing images (TSIs).**

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Jialu Li, Lingyu Sun, Chen Wu, and Liangpei Zhang are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430072, China (e-mail: lijialu@whu.edu.cn; lingyu.sun@whu.edu.cn; chen.wu@whu.edu.cn; zlp62@whu.edu.cn).

Bo Du is with the School of Computer Science, the National Engineering Research Center for Multimedia Software, the Institute of Artificial Intelligence, and Hubei Key Laboratory of Multimedia and Network Communication Engineering, Wuhan University, Wuhan 430072, China, and also with the Zhongguancun Academy, Beijing 100094, China (e-mail: gunspace@163.com).

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## I. Introduction

GLOBAL environmental change is accelerating, and the Earth's surface is experiencing increasingly dynamic transformations [1], [2]. These changes not only affect natural ecosystems but also challenge sustainable development pathways [3], [4]. From urban expansion and infrastructure construction [5], [6] to land-use and land-cover (LULC) transitions [7], [8], continuous monitoring and analysis of such dynamics are essential for environmental governance [9], [10], resource management [11], [12], and disaster assessment [13], [14]. Therefore, systematically tracking Earth's surface changes is critical for understanding and responding to global environmental change.

Early studies relied on decomposition-based methods to monitor these changes, such as Landsat-based detection of trends in disturbance and recovery (LandTrendr) [15], breaks for additive seasonal and trend (BFAST) [16], and continuous change detection and classification (CCDC) [17]. These methods analyzed temporal trajectories by decomposing long-term observations into trend, seasonal, and disturbance components. Specifically, LandTrendr detected interannual changes through temporal segmentation and piecewise linear fitting [18], BFAST identified abrupt changes by locating breakpoints in trend and seasonal signals [19], while CCDC enabled intra-annual monitoring through the integration of time-series regression and classification strategies [20]. Although these methods have achieved promising results in long-term change monitoring, their performance largely depends on predefined temporal models. As a result, they often struggle to characterize the irregular, nonlinear, and multiscale change patterns commonly observed in real-world remote sensing data. This limitation restricts their flexibility and reduces their ability to adapt to complex surface dynamics.

Recently, deep learning has become a dominant paradigm for remote sensing image analysis due to its strong capability in automatic feature learning and nonlinear representation. By effectively modeling complex and multiscale variations, deep learning methods have significantly improved change detection performance in remote sensing images [21], [22]. Representative studies include RACDNet [23], which performs arbitrary-resolution change detection through joint super-resolution reconstruction and change analysis; DGKAN [24] and CFRL [25], which improve cross-sensor change detection via structural commonality learning and cross-modal

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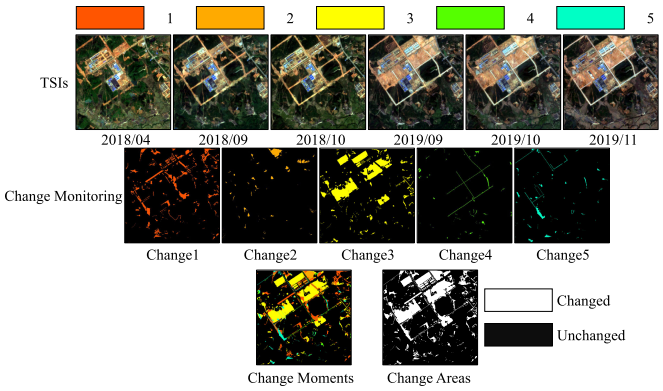



Fig. 1. Difference between change area detection, change moment identification and continuous change monitoring. White represents changed locations, black represents unchanged locations, while different colors represent changes occurring at different time points.

feature representation learning, respectively; and ARCD [26], which enhances hyperspectral change detection under limited supervision by integrating graph convolutional networks, difference-enhanced feature fusion, and progressive feature reconstruction. Meanwhile, brain-inspired spiking neural networks (SNNs) have also been explored for remote sensing tasks such as image super-resolution [27], online adaptation [28], and spatiotemporal representation learning [29]. Despite these advances, most existing methods remain focused on bitemporal or task-specific settings, making it difficult to fully exploit the temporal evolution information contained in time-series remote sensing images (TSIs). As a result, continuous surface dynamics and complex change processes are still insufficiently modeled.

With the rapid growth of computational resources, TSIs can now be processed as unified inputs for end-to-end learning and inference, which has greatly advanced the development of time-series change detection (TSCD) [30], [31]. Consequently, TSCD has attracted increasing attention in recent years [32]. TSCD mainly involves three tasks: change area detection, change moment identification, and continuous change monitoring. As illustrated in Fig. 1, all three tasks take a sequence of co-registered TSIs as input, while their outputs differ according to the target objective. Change area detection focuses on identifying regions where land-cover or object changes have occurred during the observation period. The output is typically a binary change map, in which each pixel is labeled as either changed or unchanged. Change moment identification aims to determine the temporal location of change for each pixel. The output is a multiclass map, where each pixel is assigned the time step corresponding to its detected change event. Continuous change monitoring provides a more comprehensive description of temporal evolution by detecting changes between every pair of consecutive observations. Instead of generating a single result, it produces a series of binary change maps that characterize the spatial distribution of changes throughout the entire observation period. In each map, pixels are categorized as either changed or unchanged. Overall, these three tasks can be regarded as different levels of temporal detail within TSCD. Continuous change monitoring offers the most complete representation of change evolution over time. Change moment identification summarizes this process by locating the timing of change events, whereas change area detection further simplifies the problem to a binary distinction between changed and unchanged regions.

For change area detection, early studies mainly relied on recurrent neural networks (RNNs) [33], [34], [35] and 3-D convolutional neural networks (3D CNNs) [36] to capture temporal dependencies from multitemporal observations. For instance, Meshkini proposed an 3-D CNN framework [37] for unsupervised change area detection, L-UNet [38] integrated U-Net with convolutional long short-term memory (LSTM), MC$^2$ABNet [39] employed bidirectional convolutional LSTM for temporal modeling, and UTRNet [40] introduced a time-guided LSTM to alleviate inter- and intra-annual pseudo changes. More recently, HiSTENet [41] incorporated enhanced state space models (SSMs) to better characterize long-range spatiotemporal dependencies. Although these methods extend conventional bi-temporal change detection by exploiting temporal observations over longer periods, their primary objective remains binary change area detection and determining whether a pixel has changed during the observation period. Such methods can effectively locate changed regions, but they provide limited information about the temporal dynamics of change. In many practical applications, knowing when a change occurs is equally important, since change timing offers richer temporal cues and enables a more comprehensive understanding of surface evolution processes.

For change moment identification, Multi-RLD-Net [42] was among the first methods to explicitly introduce change moment prediction into TSCD. It employed multiscale LSTMs to enhance temporal feature learning and generate moment-level change maps. Building upon this idea, CAIM-Net [43] jointly modeled change area detection and change moment identification, improving the consistency between spatial change regions and temporal change cues. More recently, AnyTime-CD [44] introduced a self-supervised framework that identifies change moments through joint temporal dynamics modeling, spectral consistency learning, and spatial alignment, without requiring manual annotations. Despite their promising performance, these methods mainly focus on identifying a single representative change moment for each pixel, typically corresponding to the most recent change event. As a result, they are unable to continuously characterize multiple changes occurring at the same location over time, limiting their ability to capture complex temporal evolution patterns.

For continuous change monitoring, existing deep learning methods can generally be categorized into three groups. The first performs semantic segmentation on each temporal image independently and then detects changes through postclassification comparison between adjacent observations [45], [46], [47]. The second extends conventional bi-temporal change detection by repeatedly applying change detection models to consecutive image pairs [48], [49]. The third processes the entire TSIs as a unified input, jointly modeling all temporal observations before identifying changes through comparisons between adjacent time steps [50].

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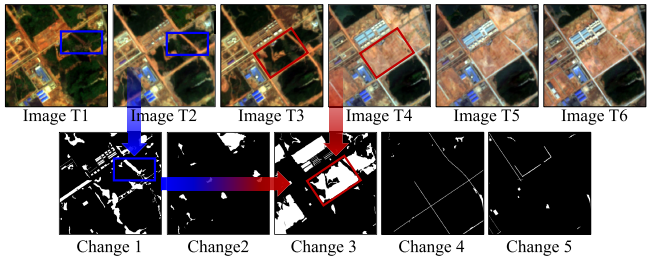



Fig. 2. Temporally asynchronous and spatially heterogeneous in time-series images.

Although the first two categories can capture local temporal variations, they treat each observation or image pair largely independently and therefore fail to fully exploit long-range temporal dependencies. The third category incorporates global temporal information, but its reliance on post-classification comparison makes the final results highly dependent on segmentation accuracy, leading to error propagation across time. Consequently, effectively modeling temporal dynamics and directly learning change evolution patterns remain key challenges for continuous change monitoring in TSIs.

Additionally, continuous changes in TSIs often exhibit temporal asynchrony and spatial heterogeneity. As illustrated in Fig. 2, similar change patterns may emerge in geographically distant regions and at different time points, making reliable spatiotemporal correspondence modeling challenging.

Existing TSCD methods mainly rely on LSTM, Transformer, or SSM architectures to model spatiotemporal dependencies. LSTM approaches process features sequentially and mainly emphasize temporal transitions, limiting their ability to associate similar changes across space and time. Transformer methods capture global dependencies effectively, but their computational and memory costs grow rapidly with spatial resolution and sequence length. Recent SSM methods improve efficiency in long-sequence modeling; however, they primarily focus ON state transitions and temporal dependency learning, providing limited capability for explicit spatiotemporal correspondence modeling. Consequently, efficiently capturing long-range spatiotemporal correlations while maintaining computational efficiency remains a fundamental challenge in TSCD.

To address this issue, we introduce a spatial-spatiotemporal LSTM (SSL) module. Unlike conventional recurrent networks that mainly emphasize temporal evolution and attention-based models that depend on pairwise global interactions, SSL jointly captures spatial and spatiotemporal correlations through dedicated scanning strategies. By integrating the input-sensitive gating mechanism of scalar LSTM (sLSTM) with the matrix-memory representation of matrix LSTM (mLSTM), SSL effectively associates similar change responses across different locations and time steps while maintaining computational efficiency.

Building upon this design, we propose a continuous change monitoring network (CCM-Net) for TSIs. CCM-Net integrates continuous change monitoring and change moment identification within a unified framework, enabling both tasks to share spatiotemporal representations and benefit from complementary learning. By jointly modeling change evolution and change timing, the proposed framework provides a more comprehensive characterization of land surface dynamics.

The main contributions are summarized as follows.

1) To bridge the methodological gap in deep learning-based continuous change monitoring, this article proposes a novel approach CCM-Net, enabling not only the detection of change occurrences but also their dynamic evolution.
2) CCM-Net introduces an SSL module to capture spatial and spatiotemporal correlations, aiming to model similar change patterns across different spatial locations and time points.
3) CCM-Net introduces a monitoring inference moment (MIM) module to infer change moments from continuous change monitoring results, integrating both tasks into a unified framework, thereby facilitating effective information exchange and enabling complementary learning.
4) The proposed CCM-Net is qualitatively and quantitatively evaluated on two representative TSIs datasets containing continuous change and change moment annotations. Experimental results show that CCM-Net improves the $F1$ score by about 2.84% or higher over state-of-the-art approaches.

The rest sections of this article are organized as follows: Section II offers a detailed elaboration of the proposed CCM-Net framework. Section III elaborates on experimental results and analysis. Section IV provides a discussion about proposed CCM-Net. Finally, we draw a conclusion in Section V.

## II. Methodology

The overall architecture of CCM-Net is illustrated in Fig. 3. The framework is designed to jointly perform continuous change monitoring and change moment identification from TSIs. First, a lightweight three-level U-Net extracts spatial features from each temporal observation. Difference features are then constructed between adjacent observations to highlight temporal variations while suppressing static background information. To enhance the representation of these differences, an SSL module is embedded at each stage of the encoder. The module employs distinct scanning strategies to capture spatial and spatiotemporal dependencies from the difference features. Specifically, sLSTM is utilized to improve the sensitivity to changes, while mLSTM is adopted to enhance the model's ability to retain information over time. The enhanced difference features are subsequently fed into the decoder to recover the original spatial resolution and generate continuous change monitoring maps at multiple stages. This multistage supervision facilitates more accurate and stable change monitoring. To further infer change moments, an MIM module is introduced at the final stage. The module analyzes the predicted change and no-change probabilities between adjacent observations and converts them into a unified change-moment representation. As a result, multiple changes can be visualized within a single

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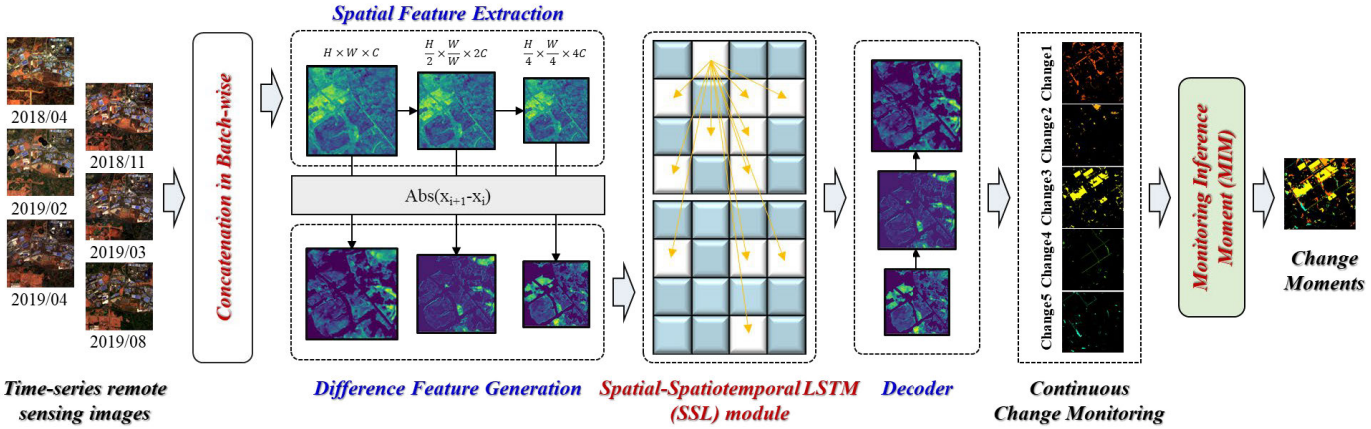



Fig. 3. Overview of the proposed CCM-Net framework, which integrates continuous change monitoring and change moment identification into a unified framework. The SSL module captures spatial and spatiotemporal dependencies, while the MIM module infers change moments from continuous change monitoring results.

map. Moreover, the proposed design establishes a direct link between continuous change monitoring and change moment identification, enabling joint optimization of the two tasks and providing a more comprehensive characterization of temporal change dynamics in TSIs.

### A. Difference Feature Generation

In CCM-Net, a lightweight three-level U-Net is employed to extract spatial features from TSIs. To enable efficient processing of multitemporal observations, the temporal and batch dimensions are merged into a unified batch dimension, resulting in feature tensors of size $[T \times B, C, H, W]$. This strategy avoids repeated feature extraction across time steps and allows all observations to be processed in parallel, thereby improving computational efficiency during training and inference.

Each stage of the U-Net consists of two convolutional layers with $3 \times 3$ kernels, a stride of 1, and a padding of 1. A sigmoid linear unit (SiLU) activation function follows each convolution to enhance the nonlinearity of feature representations. For downsampling, a convolutional layer with a $4 \times 4$ kernel and a stride of 2 is adopted instead of max pooling. By introducing learnable parameters into the downsampling process, the network can adaptively preserve more informative spatial features and improve feature representation capability.

Continuous change monitoring aims to capture temporal variations between adjacent observations. After spatial feature extraction, the feature tensors are reshaped from $[T \times B, C, H, W]$ to $[T, B, C, H, W]$. Difference features are then constructed by computing the absolute differences between consecutive temporal features. Compared with directly using original features, this operation explicitly highlights changed regions while suppressing static background responses, making subsequent temporal modeling more focused on change-related information. The procedure is performed at every U-Net stage, producing multiscale temporal difference features for continuous change monitoring. As illustrated in Fig. 4, the temporal difference features are computed as

$$D_i(j) = |Conv_i(j+1) - Conv_i(j)| \tag{1}$$

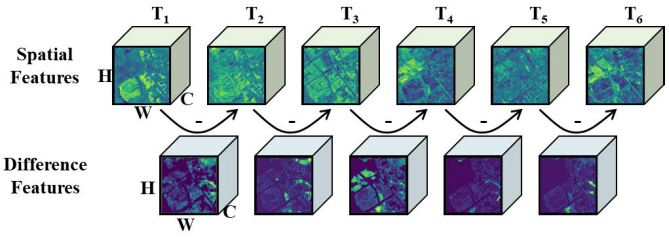



Fig. 4. Flowchart of the difference extraction.

where $D_i(j)$ denotes the difference feature at the $i$th U-Net stage and the $j$th temporal interval, $Conv_i$ represents the convolution layers used for spatial feature extraction at the $i$th stage, and $T$ is the number of temporal observations. The temporal index satisfies $1 \le j \le T-1$.

### B. SSL Module

*1) sLSTM and mLSTM:* To overcome the scalability limitations of conventional LSTM, two enhanced variants, sLSTM and mLSTM, were proposed in xLSTM [51]. This section briefly reviews the standard LSTM and highlights the key differences introduced by sLSTM and mLSTM.

Standard LSTM employs a memory cell to capture long-range dependencies in sequential data. Information flow is regulated by three gating mechanisms, including the input gate, forget gate, and output gate, which jointly control memory updating and information retention. The memory cell at time step $t$ is formulated as follows:

$$\begin{aligned} c_t &= f_t c_{t-1} + i_t z_t \\ h_t &= o_t \cdot \varphi(c_t) \\ z_t &= \varphi\left(w_z^T x_t + r_z h_{t-1} + b_z\right) \\ i_t &= \sigma\left(w_i^T x_t + r_i h_{t-1} + b_i\right) \\ f_t &= \sigma\left(w_f^T x_t + r_f h_{t-1} + b_f\right) \\ o_t &= \sigma\left(w_o^T x_t + r_o h_{t-1} + b_o\right). \end{aligned} \tag{2}$$

In the above equations, $c, i, f, o$ denote the cell state, input gate, forget gate, and output gate, respectively. $h$ represents the

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hidden state, while $z$ denotes the cell input. The parameters $w, r, b$ correspond to the input weight, recurrent weight, and bias term of each gate. The functions $\varphi$ and $\sigma$ denote the hyperbolic tangent (tanh) and sigmoid activation functions, respectively.

Compared with standard LSTM, sLSTM introduces two major modifications. First, the sigmoid activations in the input and forget gates are replaced with exponential functions, which alleviate gradient attenuation and improve information propagation over long sequences. Second, an additional normalization state is incorporated into the memory update process to stabilize cell-state dynamics and reduce sensitivity to input fluctuations. The resulting formulations are given below, where the modified components are highlighted in bold

$$\begin{aligned} c_t &= f_t c_{t-1} + i_t z_t \\ h_t &= o_t \cdot \left(\frac{\boldsymbol{c_t}}{\boldsymbol{n_t}}\right), \quad \boldsymbol{n_t} = \boldsymbol{f_t n_{t-1}} + \boldsymbol{i_t} \\ z_t &= \varphi\left(w_z^T x_t + r_z h_{t-1} + b_z\right) \\ i_t &= \boldsymbol{exp}\left(w_i^T x_t + r_i h_{t-1} + b_i\right) \\ f_t &= \boldsymbol{exp}\left(w_f^T x_t + r_f h_{t-1} + b_f\right) \\ o_t &= \sigma\left(w_o^T x_t + r_o h_{t-1} + b_o\right). \end{aligned} \tag{3}$$

In these equations, all nonbolded variables retain the same definitions as those in standard LSTM. The bolded terms indicate the newly introduced components. Specifically, $\boldsymbol{exp}$ denotes the exponential activation function, and $\boldsymbol{n_t}$ represents an additional normalization state used to regulate memory updates and enhance numerical stability.

Building upon sLSTM, mLSTM further extends the memory representation from a scalar state to a matrix state. Instead of directly updating the memory cell through element-wise operations, mLSTM adopts a query-key-value (QKV) mechanism to construct and update the memory matrix. This design is inspired by the similarity computation strategy used in Transformer architecture and enables richer interactions among feature dimensions. The corresponding formulations are presented below, with the modifications relative to sLSTM highlighted in bold

$$\begin{aligned} \boldsymbol{C}_t &= f_t \boldsymbol{C}_{t-1} + i_t \boldsymbol{v}_t \boldsymbol{k}_t^T \\ h_t &= o_t \odot \left(\boldsymbol{C}_t \frac{\boldsymbol{q}_t}{\max\left\{\left|\boldsymbol{n}_t^T \boldsymbol{q}_t\right|, 1\right\}}\right) \\ n_t &= f_t n_{t-1} + \boldsymbol{k}_t \\ \boldsymbol{q}_t &= \boldsymbol{W}_q \boldsymbol{x}_t + \boldsymbol{b}_q \\ \boldsymbol{k}_t &= \frac{1}{\sqrt{d}} \boldsymbol{W}_k \boldsymbol{x}_t + \boldsymbol{b}_k \\ \boldsymbol{v}_t &= \boldsymbol{W}_v \boldsymbol{x}_t + \boldsymbol{b}_v \\ i_t &= \exp\left(w_i^T x_t + b_i\right) \\ f_t &= \exp\left(w_f^T x_t + b_f\right) \\ o_t &= \sigma\left(\boldsymbol{W}_o x_t + \boldsymbol{b}_o\right). \end{aligned} \tag{4}$$

In the mLSTM formulation, the nonbolded terms retain the same meanings as those in sLSTM. The bolded variables denote the newly introduced matrix-based components. In particular, $\boldsymbol{q}$, $\boldsymbol{k}$, $\boldsymbol{v}$ represent the query, key, and value states, respectively, while $\boldsymbol{C}$ denotes the matrix memory state. In addition, $\boldsymbol{W_o}$ and $\boldsymbol{b_o}$ are extended from vector parameters to matrix parameters, allowing the model to capture more complex dependencies and improve long-range information modeling.

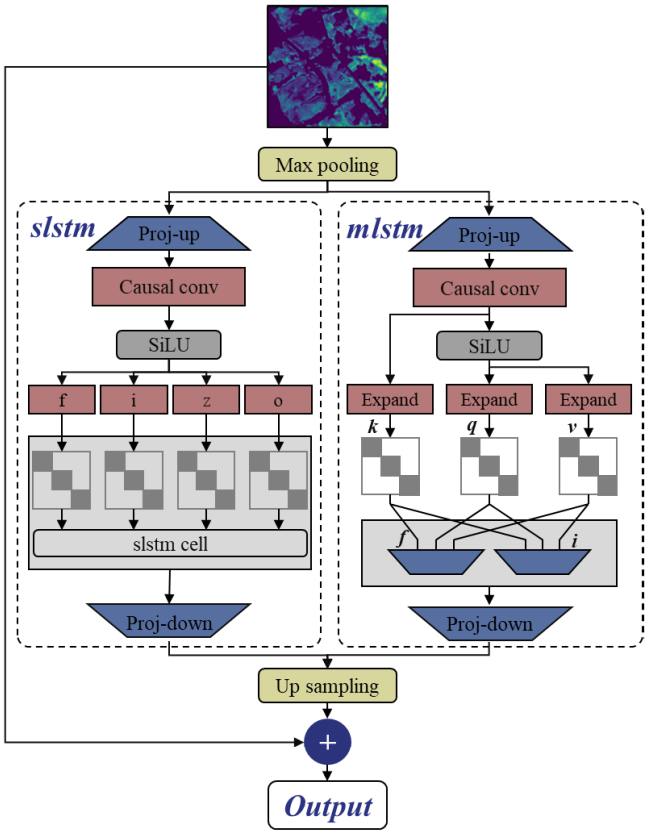



Fig. 5. Structure of the proposed SSL module. It integrates sLSTM and mLSTM for spatial and spatiotemporal correlation modeling.

*2) SSL Module:* Building on sLSTM and mLSTM, we propose an SSL module to capture spatial correlations within individual difference features and spatiotemporal correlations across time-series difference features. By combining the input-sensitive gating mechanism of sLSTM with the matrix-memory representation of mLSTM, SSL enhances responsiveness to subtle changes while improving long-range dependency modeling.

The architecture of SSL is illustrated in Fig. 5. Since the module is embedded at each U-Net stage, a Max Pooling layer with a kernel size and stride of 2 is applied before SSL to reduce the spatial resolution of shallow features and improve computational efficiency.

SSL consists of two parallel branches. In the first branch, the input features are projected into a higher-dimensional space, where nonlinear feature distributions can be represented more effectively. A causal convolution is then applied to preserve the sequential order, followed by a SiLU activation. The resulting features are processed by an sLSTM, which employs exponential gating to improve sensitivity to input variations and mitigate gradient attenuation. In the second branch, the features undergo the same projection and causal convolution

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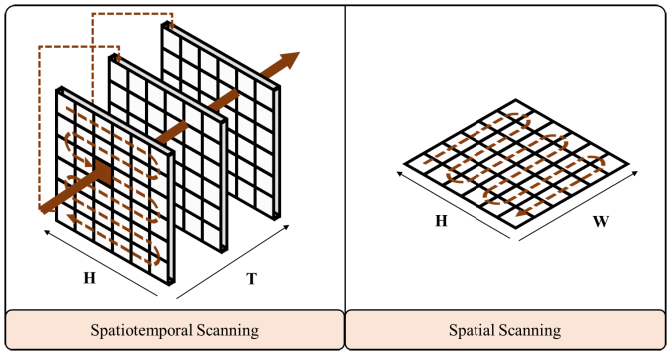



Fig. 6. Spatial and spatiotemporal scanning strategies in SSL. The spatiotemporal scan associates similar change patterns across space and time, while the spatial scan strengthens spatial consistency.

operations before being fed into an mLSTM. By extending the memory state from a vector to a matrix, mLSTM captures richer feature interactions and long-range dependencies. However, applying mLSTM directly to high-resolution features incurs substantial computational cost. Therefore, the query and key representations are generated only at the deepest U-Net stage and reused in shallower stages through upsampling, enabling efficient cross-stage correlation modeling. Finally, the outputs of the two branches are fused with the input difference features to obtain correlation-enhanced representations.

Compared with the original sLSTM and mLSTM architectures, the proposed SSL module is specifically designed for TSCD. It separates spatial and spatiotemporal correlation modeling into two parallel branches and introduces dedicated scanning strategies to propagate information across both spatial locations and temporal observations. In addition, the cross-stage reuse of low-resolution query and key representations significantly reduces computational complexity while preserving the long-range modeling capability of mLSTM.

To model the asynchronous and spatially heterogeneous changes in TSIs, SSL employs two scanning strategies, as illustrated in Fig. 6. For spatiotemporal correlation modeling, the difference features are scanned along the temporal dimension at each spatial location, while a row-wise scan is simultaneously performed within each feature map from the top-left to the bottom-right direction. The hidden state is therefore updated using information from both temporal observations and neighboring spatial positions, enabling the association of similar change patterns across space and time. For spatial correlation modeling, the same scanning mechanism is applied within a single-time difference feature, strengthening spatial consistency and improving the representation of changed regions.

### C. MIM Module

To identify change moments from continuous change monitoring results, CCM-Net incorporates a MIM module at the final stage. As illustrated in Fig. 7, MIM aggregates the predicted change probabilities across all consecutive observations and estimates the probability distribution of change moments. Let $P_t^c$ and $P_t^{nc}$ represent the predicted change and no-change probabilities between the $t$th and $(t+1)$th

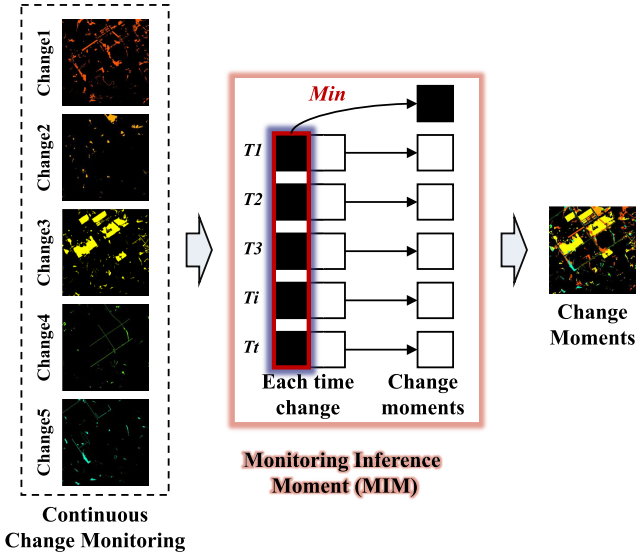



Fig. 7. Structure of the MIM module. It aggregates continuous change monitoring results to infer change moment.

observations, respectively, where $t = 1, \ldots, T-1$. Because a change moment is associated with the transition between two adjacent observations, the change probability of each temporal interval serves as the initial confidence for the corresponding change moment

$$M_t = P_t^c, \quad t = 1, \ldots, T-1. \tag{5}$$

To characterize the probability that a pixel remains unchanged during the entire observation period, the minimum no-change probability among all temporal intervals is taken as the confidence of the unchanged class

$$M_0 = Min\left(P_t^{nc}\right). \tag{6}$$

The unchanged confidence and the inferred change moment probabilities are then concatenated to construct the change moment probability vector. A Softmax operation is subsequently applied to obtain a normalized probability distribution. The final change moment probability vector is expressed as

$$\mathbf{M} = f[M_0, M_1, \ldots, M_{T-1}] \tag{7}$$

where $M_t (t = 1, \ldots, T-1)$ denotes the confidence that a change occurs at the corresponding temporal interval, $M_0$ represents the confidence of the unchanged class, and $M$ denotes the final probability distribution of change moments. The $f$ refers to the Softmax activation used for probability normalization.

By integrating continuous change predictions across all temporal intervals, MIM module transforms change monitoring results into a unified change moment representation. This design establishes a direct connection between continuous change monitoring and change moment identification, allowing the two tasks to reinforce each other and improving the consistency of spatiotemporal change analysis.

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### D. Loss Function and Optimization

To address the imbalance between change and no-change samples as well as between easy and hard examples, we integrate the focal-weighted cross-entropy loss (FWCL) into the CCM-Net framework for more effective network optimization. The FWCL is defined as follows:

$$L_f = -\frac{1}{N}\sum_{i=1}^{N}\left[R\,(1-\hat{y}_n)^{\gamma}\, y_n \log \hat{y}_n\right]. \tag{8}$$

We generate continuous change monitoring results at each U-Net layer and compute the loss, ensuring stable stage-wise predictions across different scales. At the final U-Net layer, we infer the change moment and calculate the corresponding loss, enabling effective information transfer between the two tasks

$$L = L_M + \frac{L_{D1}}{\text{len}(TSIs)-1} + \frac{L_{D2}}{\text{len}(TSIs)-1} + \frac{L_{D3}}{\text{len}(TSIs)-1}. \tag{9}$$

In above formulations, $L$ represents the overall loss used for model optimization. Specifically, $L_M$ denotes the change moment loss, while $L_{D1}$, $L_{D2}$, and $L_{D3}$ are the continuous change monitoring losses computed at each U-Net stage. At each stage, the model outputs $len(TSIs) - 1$ change maps. Therefore, $L_{D1}$, $L_{D2}$, and $L_{D3}$ are averaged over this length to ensure consistent scaling. All losses $L_M$, $L_{D1}$, $L_{D2}$, and $L_{D3}$ are calculated based on $L_f$, which refers to the FWCL loss. In this context, $N$ denotes the total number of training samples, and $R$ represents the ratio of the number of training samples excluding the current class samples to all training samples. The predicted value is denoted as $\hat{y}_n$, and the corresponding ground truth label as $y_n$. The focusing parameter $\gamma$ is a positive scalar that adjusts the weight of easy samples. To maintain consistency with the original FWCL formulation [40], we set $\gamma = 2$ in the proposed CCM-Net framework.

## III. Experimental Results and Analysis

### A. Experimental Setup

*1) Datasets:* We evaluate our method on two widely used TSCD datasets: DynamicEarthNet and SpaceNet7, both providing long-term multitemporal satellite images. We adopt the same data partitioning strategies as in prior studies to ensure a fair comparison. Representative TSIs together with their corresponding continuous change maps and change moment annotations from both datasets are shown in Figs. 8 and 9.

DynamicEarthNet [52] is the Daily Multispectral Satellite Semantic Change Segmentation dataset, which is a globally distributed benchmark for semantic change analysis. It comprises 55 areas of interest (AOIs) annotated with pixel-wise LULC labels. The dataset provides multiresolution Sentinel-2 imagery at 10, 20, and 60 m spatial resolutions, acquired from January 2018 to December 2019. Due to the absence of strict cloud-filtering during data collection, a considerable portion of the images is affected by cloud contamination. To ensure data reliability, cloud-contaminated images were excluded during preprocessing. After quality control, 30 labeled AOIs were retained, each containing six temporally ordered images. For model development, four spectral bands with 10 m spatial resolution—blue (B2), green (B3), red (B4), and near-i nfrared (B8)—were selected. The original images, with a spatial size of 1024 × 1024 pixels, were partitioned into 64 × 64 patches for training and inference. The processed dataset was divided into training, validation, and testing subsets following an 8:1:1 ratio. Owing to its global coverage and temporally dense observations, DynamicEarthNet provides a challenging benchmark for assessing the robustness and generalization capability of the proposed method.

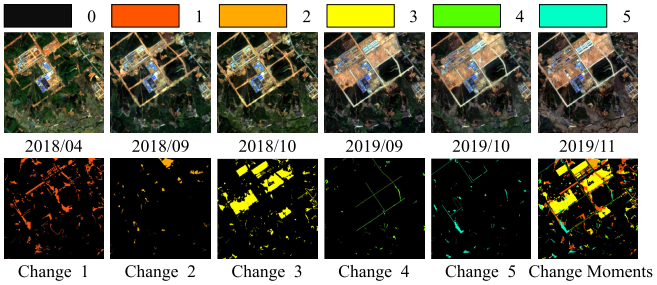



Fig. 8. Representative images from the DynamicEarthNet dataset, along with their corresponding continuous change monitoring labels and change moment labels. In the corresponding label maps, black represents unchanged areas, while different colors represents changes occurring at different time points.

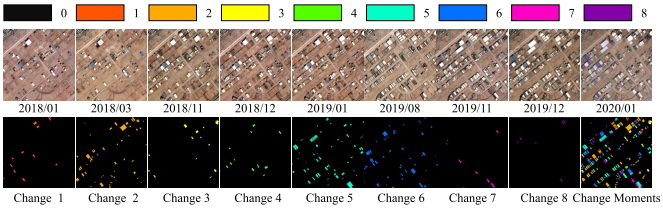



Fig. 9. Representative images from the SpaceNet7 dataset, along with their corresponding continuous change monitoring labels and change moment labels. In the corresponding label maps, black represents unchanged areas, while different colors represents changes occurring at different time points.

In the DynamicEarthNet dataset, each image is provided with pixel-wise LULC annotations. To construct continuous change monitoring labels and corresponding change moment labels, we analyze temporal transitions in these semantic annotations. Specifically, semantic labels from consecutive time steps are compared to identify pixel-wise class transitions, thereby generating a sequence of binary change maps that describe continuous changes across TSIs. Based on these maps, change moment labels are further derived. For each pixel, a label of "0" denotes no change throughout the entire sequence. If one or more changes are detected, the pixel is assigned the time index corresponding to the most recent transition. For example, a label of "1" indicates a change between the first and second images, while "2" denotes a change between the second and third images.

SpaceNet7 [53] is the Multitemporal Urban Development SpaceNet (SpaceNet7) dataset, comprising 60 AOIs annotated with building semantic segmentation labels. The images, collected from July 2017 to January 2020, have a spatial resolution of 4 m and contain four spectral bands—red, green,

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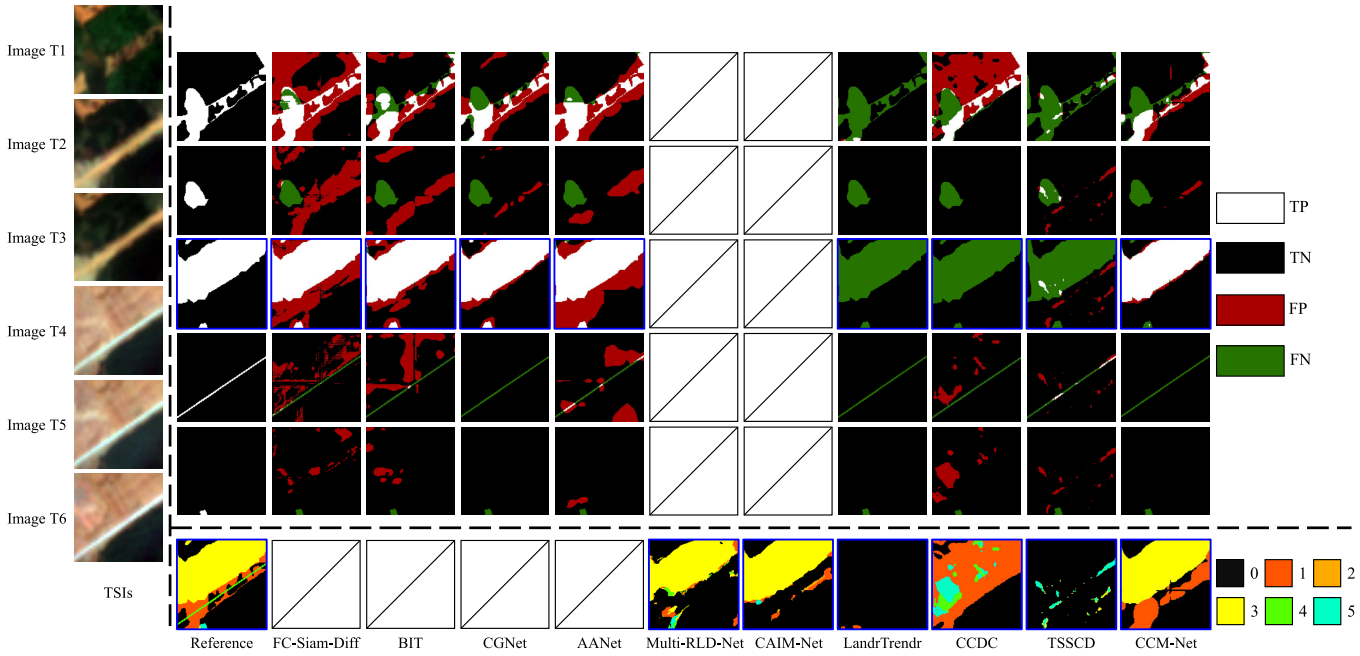



Fig. 10. Qualitative comparison of CCM-Net and nine baselines on DynamicEarthNet. False positives (erroneous changed) are highlighted in red and false negatives (erroneous unchanged) in green.

blue, and near-infrared—with an original size of $1024 \times 1024$ pixels. For model training, each image was further cropped into $64 \times 64$ pixel patches. Images with significant cloud cover were excluded to maintain data quality, although cloud influence in SpaceNet7 is generally less pronounced than in DynamicEarthNet. Ultimately, the dataset contains 60 labeled AOIs, each comprising nine TSIs, which were split into training, validation, and testing subsets using an 8:1:1 ratio. Continuous change and change moment labels were generated following the same procedure as for the DynamicEarthNet dataset. Given its coverage of variations in building density and distribution, SpaceNet7 provides a suitable benchmark for evaluating the robustness of the proposed method.

*2) Evaluation Protocol:* To comprehensively evaluate the proposed CCM-Net, we employ five quantitative metrics: overall accuracy (OA), $F1$-score ($F1$), mean intersection over union (mIoU), precision (Pre), and recall (Rec). The $F1$-score, which balances Precision and Recall, effectively mitigates biases arising from class imbalance. The mIoU, which is widely used in semantic and instance segmentation, measures pixel-level segmentation accuracy. These metrics are particularly suitable for assessing model performance.

*3) Comparison Algorithm:* This article compares the proposed method with nine advanced approaches. First, four bi-temporal change detection methods—FC-Siam-Diff [54], BIT [55], CGNet [56], and AANet [57]—use consecutive image pairs to generate continuous changes. Second, two deep learning-based TSCD methods—Multi-RLD-Net [42] and CAIM-Net [43]—directly leverage TSIs to predict change moment maps. Third, two decomposition-based TSCD methods—LandTrendr [15] and CCDC [17]—extract both continuous change monitoring results and corresponding change moments from TSIs; although unsupervised, they are included due to the limited availability of effective continuous change monitoring methods. Finally, TSSCD [50], a deep learning-based TSCD method, jointly extracts continuous changes and change moments from TSIs.

*4) Experimentation Details:* The proposed CCM-Net framework was implemented in PyTorch and trained on an NVIDIA GeForce RTX 3090 Ti GPU. The Adam optimizer [58] was used, with initial learning rates of 0.0004 and 0.0001 for the DynamicEarthNet and SpaceNet7 datasets, respectively. The learning rate was reduced by half whenever validation performance plateaued. Training for both datasets was conducted for 50 epochs with a batch size of 64.

## B. Benchmark Comparison and Analysis

*1) Analysis for DynamicEarthNet Dataset:* Fig. 10 presents the continuous change monitoring results and change moment maps on the DynamicEarthNet test set generated by CCM-Net and nine comparison methods. Multi-RLD-Net and CAIM-Net are specifically designed for change moment identification and are therefore excluded from the continuous monitoring comparison. As illustrated by the blue-boxed region in Fig. 10, the iterative bitemporal methods—FC-Siam-Diff, BIT, CGNet, and AANet—operate only on temporally adjacent image pairs, which limits their ability to model long-term temporal dependencies and leads to increased false detections. In contrast, the TSCD methods utilize full temporal sequences; however, LandTrendr and CCDC are unsupervised pixel-based approaches, while TSSCD performs pixel-wise classification at each time step, making it prone to error accumulation. As a result, these methods struggle to achieve accurate and stable continuous change monitoring. By comparison, CCM-Net

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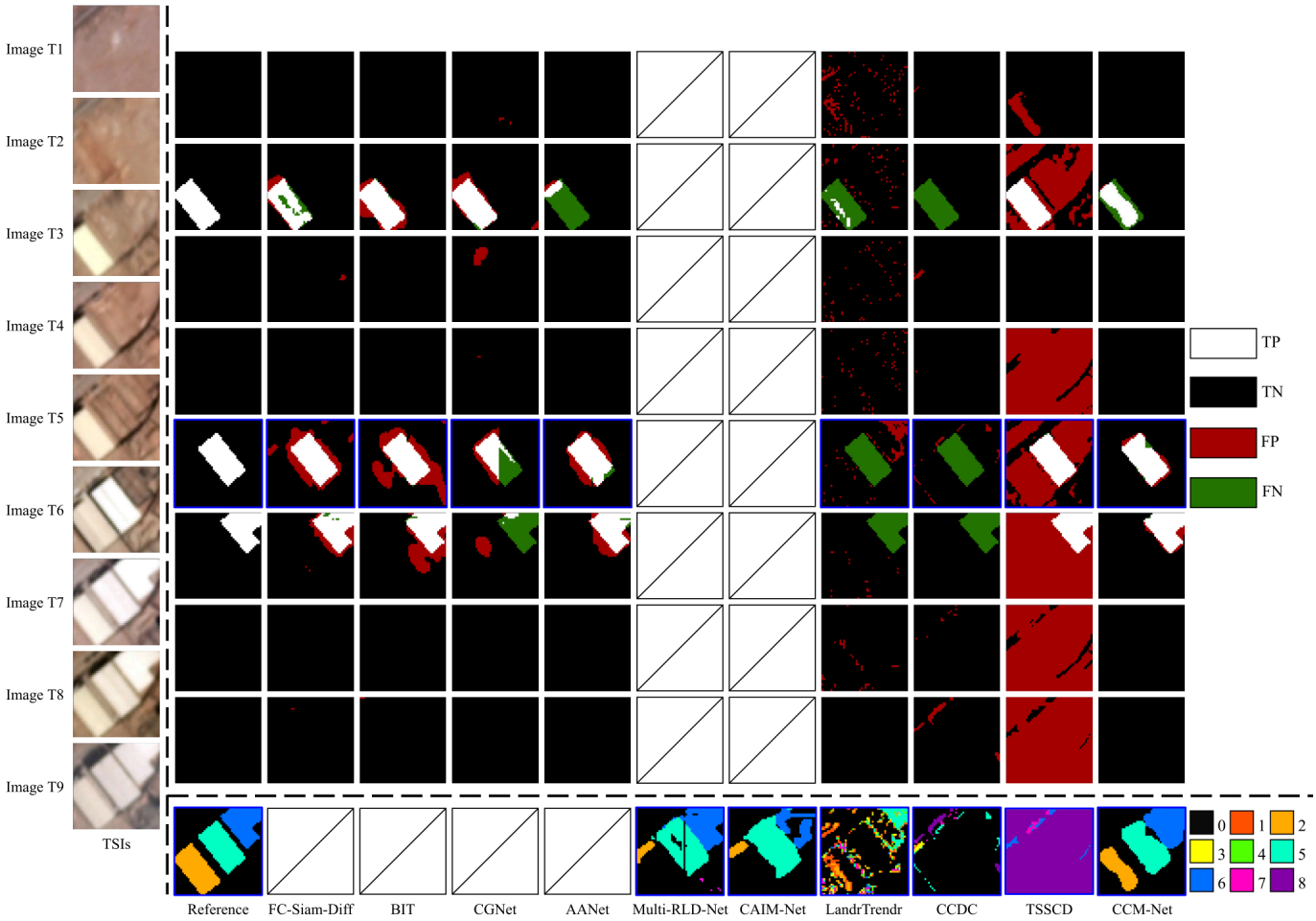



Fig. 11. Qualitative comparison of CCM-Net and nine baselines on SpaceNet7. False positives (erroneous changed) are highlighted in red and false negatives (erroneous unchanged) in green.

produces substantially fewer false positives and false negatives, demonstrating its superior capability in jointly modeling spatial and temporal dependencies within TSIs.

For change moment identification, as illustrated by the blue-boxed region in Fig. 11, the iterative bi-temporal methods are excluded because they do not explicitly estimate change moments. The TSCD methods—LandTrendr, CCDC, and TSSCD—infer change moments indirectly from monitoring results, which constrains their identification accuracy. Although Multi-RLD-Net and CAIM-Net are tailored for change moment identification, CCM-Net achieves the best overall performance by directly optimizing this task through a dedicated loss function.

Table I reports the quantitative evaluation results of CCM-Net and nine comparison methods on the DynamicEarthNet test set. The results indicate that CCM-Net achieves superior performance in both continuous change monitoring and change moment identification. This improvement is primarily due to the proposed SSL module, which captures spatiotemporal correlations, and the joint optimization strategy that explicitly models the intrinsic relationship between the two tasks within a unified framework.

*2) Analysis for SpaceNet7 Dataset:* Fig. 11 presents continuous change monitoring maps and change moment identification maps on the test set of the SpaceNet7 dataset. Compared with DynamicEarthNet, SpaceNet7 provides only building change labels and exhibits extreme class imbalance, with very few samples showing changes. This imbalance poses significant challenges for both continuous change monitoring and change moment identification tasks. In continuous change monitoring, as illustrated by the blue-boxed region in Fig. 11, bi-temporal methods such as FC-Siam-Diff and BIT tend to produce many false positives, while CGNet and AANet, benefiting from multilayer difference feature fusion, are more prone to false negatives but achieve fewer false positives overall. TSCD methods such as LandTrendr and CCDC, designed for natural land cover, perform poorly in detecting building changes and yield many false negatives. TSSCD, which classifies each image in the time series independently, suffers from misclassifications that increase false positives. In contrast, CCM-Net generates maps with substantially fewer false positives and false negatives, indicating its superior ability to capture spatiotemporal dependencies and monitor continuous changes accurately.

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TABLE I

QUANTITATIVE COMPARISON OF CCM-NET AND NINE BASELINES ON DYNAMICEARTHNET. THE TABLE HIGHLIGHTS THE HIGHEST VALUES IN RED, AND THE SECOND-HIGHEST VALUES IN BLUE

| Method | Change Monitoring | | | | | Change Moment | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | OA | F1 | mIoU | Pre | Rec | OA | F1 | mIoU | Pre | Rec |
| FC-Siam-Diff | 76.47 | 25.44 | 45.03 | 17.77 | **44.80** | / | / | / | / | / |
| BIT | 79.85 | 27.48 | 47.49 | 20.28 | **42.61** | / | / | / | / | / |
| CGNet | 86.27 | 27.17 | **50.81** | **25.89** | 28.57 | / | / | / | / | / |
| AANet | 82.76 | **28.43** | 49.36 | 22.64 | 38.22 | / | / | / | / | / |
| Multi-RLD-Net | / | / | / | / | / | 74.87 | 41.14 | 29.66 | 45.89 | **41.87** |
| CAIM-Net | / | / | / | / | / | **76.01** | **43.83** | **32.54** | **53.80** | 40.99 |
| LandTrendr | 88.94 | 6.01 | 46.00 | 12.60 | 3.94 | 65.66 | 16.15 | 12.17 | 15.90 | 16.87 |
| CCDC | **89.50** | 13.19 | 48.24 | 25.53 | 8.90 | 72.22 | 22.16 | 17.11 | 26.66 | 23.07 |
| TSSCD | 72.64 | 15.72 | 40.23 | 10.86 | 28.46 | 66.04 | 16.50 | 12.77 | 17.20 | 17.71 |
| CCM-Net | **91.36** | **35.74** | **56.45** | **53.63** | 26.80 | **76.11** | **45.64** | **32.90** | **50.44** | **42.89** |

TABLE II

QUANTITATIVE COMPARISON OF CCM-NET AND NINE BASELINES ON SPACENET7. THE TABLE HIGHLIGHTS THE HIGHEST VALUES IN RED, AND THE SECOND-HIGHEST VALUES IN BLUE

| Method | Change Monitoring | | | | | Change Moment | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | OA | F1 | mIoU | Pre | Rec | OA | F1 | mIoU | Pre | Rec |
| FC-Siam-Diff | 98.88 | 19.31 | 54.78 | 12.25 | **45.57** | / | / | / | / | / |
| BIT | 99.18 | 21.66 | 55.66 | 15.09 | 38.37 | / | / | / | / | / |
| CGNet | **99.40** | 24.10 | 56.55 | 19.28 | 32.13 | / | / | / | / | / |
| AANet | 99.39 | **28.11** | **57.87** | **21.56** | **40.37** | / | / | / | / | / |
| Multi-RLD-Net | / | / | / | / | / | 97.63 | **41.84** | **30.08** | **47.02** | **39.83** |
| CAIM-Net | / | / | / | / | / | **97.66** | 40.73 | 29.23 | 44.74 | 39.57 |
| LandTrendr | 97.44 | 0.85 | 48.93 | 0.48 | 3.72 | 79.93 | 10.38 | 9.16 | 11.14 | 12.43 |
| CCDC | 98.71 | 2.15 | 49.90 | 1.39 | 4.78 | 90.50 | 12.39 | 11.04 | 12.16 | 14.51 |
| TSSCD | 87.38 | 5.03 | 44.96 | 3.10 | 13.38 | 65.69 | 10.80 | 8.50 | 11.54 | 12.21 |
| CCM-Net | **99.55** | **30.95** | **58.93** | **28.45** | 33.94 | **98.06** | **45.64** | **34.47** | **45.86** | **45.94** |

For change moment identification, as illustrated by the blue-boxed region in Fig. 11, LandTrendr and CCDC fail to detect precise change moments due to poor continuous change monitoring performance, while TSSCD often identifies changes at the last time step, limiting accuracy. Dedicated change moment identification networks, Multi-RLD-Net and CAIM-Net, achieve higher accuracy but still miss some regions. CCM-Net produces the most precise and complete change moment maps by incorporating spatial information and explicitly modeling neighboring change regions, reducing false detections in small or isolated areas.

Table II reports the quantitative evaluation of all methods on the SpaceNet7 test set. Due to class imbalance, change monitoring accuracy on SpaceNet7 is lower than on DynamicEarthNet. Among comparison methods, CCM-Net consistently achieves the best OA, $F1$ scores, and mIoU for both continuous change monitoring and change moment identification, outperforming both traditional and dedicated methods.

In summary, CCM-Net demonstrates robust superiority on both SpaceNet7 and DynamicEarthNet datasets. Across tasks and methods, it consistently achieves the highest accuracy and most reliable visualization results, highlighting its effectiveness in jointly modeling continuous changes and precise change moments.

### C. Ablation Studies

In this section, ablation studies are conducted to investigate the contributions of the SSL module and the major components of CCM-Net to continuous change monitoring and change moment identification. The results are summarized in Tables III and IV.

*1) SSL Module for Spatial and Spatiotemporal Correlation Computation:* To assess the effectiveness of the SSL module and its two branches—sLSTM and mLSTM—across spatial and spatiotemporal dimensions, we conduct an ablation study on the DynamicEarthNet and SpaceNet7 datasets, summarized in Table III. On DynamicEarthNet, using only the Encoder (U-Net) yields limited performance in both continuous change monitoring and change moment identification. Introducing sLSTM primarily improves $F1$ scores by capturing subtle spatial changes, while mLSTM enhances OA by preserving more information. Integrating both sLSTM and mLSTM in the SSL module achieves a balanced improvement across OA and $F1$ scores. The full CCM-Net model, leveraging the

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TABLE III

SSL MODULE ABLATION RESULTS ON BOTH DATASETS. HERE, S-S MEANS SPATIAL SLSTM, S-M MEANS SPATIAL MLSTM, S-SSL MEANS SPATIAL SSL MODULE, ST-S MEANS SPATIOTEMPORAL SLSTM, ST-M MEANS SPATIOTEMPORAL MLSTM, AND ST-SSL MEANS SPATIOTEMPORAL SSL MODULE

| | **DynamicEarthNet** | | | | | | **SpaceNet7** | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Change monitoring** | | | **Change Moment** | | | **Change monitoring** | | | **Change Moment** | | |
| | **OA** | **F1** | **mIoU** | **OA** | **F1** | **mIoU** | **OA** | **F1** | **mIoU** | **OA** | **F1** | **mIoU** |
| Encoder | 90.22 | 30.80 | 54.10 | 72.13 | 43.50 | 31.05 | 99.36 | 25.17 | 56.87 | 97.35 | 41.15 | 31.02 |
| S-S | 90.41 | 31.62 | 54.49 | 73.53 | 44.22 | 31.88 | 99.49 | 29.90 | 58.53 | 97.52 | 43.88 | 32.95 |
| S-M | **91.56** | 32.42 | 55.37 | **76.70** | 42.97 | 30.96 | 99.60 | 28.81 | 58.21 | 98.15 | 42.56 | 32.16 |
| S-SSL | 90.55 | 31.92 | 54.67 | 74.96 | 45.75 | 32.90 | 99.58 | 29.50 | 58.44 | 97.98 | 43.34 | 32.66 |
| ST-S | 89.93 | 33.14 | 54.77 | 72.64 | 45.19 | 32.26 | 99.46 | 29.98 | 58.55 | 97.54 | 44.37 | 33.31 |
| ST-M | 91.34 | 31.17 | 54.86 | 75.45 | 41.17 | 29.64 | **99.61** | 27.27 | 57.70 | **98.26** | 41.70 | 31.50 |
| ST-SSL | 90.98 | 32.60 | 55.13 | 75.66 | 45.10 | 32.42 | 99.51 | 26.99 | 57.55 | 97.78 | 42.06 | 31.73 |
| CCM-Net | 91.36 | **35.74** | **56.45** | 76.11 | **45.64** | **32.90** | 99.55 | **30.95** | **58.93** | 98.06 | **45.64** | **34.47** |

TABLE IV

LOSS ABLATION RESULTS ON BOTH DATASETS. HERE, SSL MEANS SPATIAL-SPATIOTEMPORAL LSTM MODULE, DEEP-SUP MEANS DEEP SUPERVISION ON CHANGE MONITORING, AND MIM MEANS MONITORING INFERENCE MOMENT MODULE

| | | | **DynamicEarthNet** | | | | | | **SpaceNet7** | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | **Change Monitoring** | | | **Change Moment** | | | **Change Monitoring** | | | **Change Moment** | | |
| **SSL** | **Deep-Sup** | **MIM** | **OA** | **F1** | **mIoU** | **OA** | **F1** | **mIoU** | **OA** | **F1** | **mIoU** | **OA** | **F1** | **mIoU** |
| | | | 75.16 | 24.16 | 43.94 | / | / | / | 99.18 | 18.93 | 54.82 | / | / | / |
| ✓ | | | 80.94 | 25.55 | 47.47 | / | / | / | 99.03 | 20.31 | 55.17 | / | / | / |
| | ✓ | | 82.48 | 29.96 | 49.71 | / | / | / | 99.30 | 22.40 | 55.95 | / | / | / |
| | | ✓ | 90.81 | 28.96 | 53.78 | 74.93 | 43.20 | 30.93 | 99.34 | 24.93 | 56.79 | 97.25 | 40.24 | 30.34 |
| ✓ | ✓ | | 84.22 | 30.70 | 50.89 | / | / | / | 99.17 | 24.31 | 56.50 | / | / | / |
| ✓ | | ✓ | 90.36 | 33.94 | 55.28 | 74.59 | 45.22 | 32.32 | 99.52 | 26.67 | 57.45 | 97.87 | 42.02 | 31.64 |
| | ✓ | ✓ | 90.22 | 30.80 | 54.10 | 72.13 | 43.50 | 31.05 | 99.36 | 25.17 | 56.87 | 97.35 | 41.15 | 31.02 |
| ✓ | ✓ | ✓ | **91.36** | **35.74** | **56.45** | **76.11** | **45.64** | **32.90** | **99.55** | **30.95** | **58.93** | **98.06** | **45.64** | **34.47** |

SSL module across both spatial and spatiotemporal domains, achieves the best overall performance.

On SpaceNet7, similar trends are observed: the SSL module consistently improves both tasks, and CCM-Net achieves the highest performance, demonstrating the generalizability of the approach. Overall, the ablation study indicates that sLSTM contributes more to $F1$ score enhancement, mLSTM primarily improves OA, and their joint modeling in CCM-Net effectively balances the two metrics, leading to superior performance across datasets and tasks.

*2) Major Components of CCM-Net:* CCM-Net consists of three key components: SSL module, deep supervision strategy, and MIM module. To evaluate the contribution of each component, ablation experiments are conducted on the DynamicEarthNet and SpaceNet7 datasets, and the results are summarized in Table IV.

On DynamicEarthNet, the baseline model without SSL, Deep-Sup, or MIM achieves relatively limited performance. Introducing SSL alone improves both OA and $F1$ scores, indicating that spatial and spatiotemporal correlation modeling is beneficial for continuous change monitoring. Deep-Sup provides larger improvements in $F1$ score, demonstrating the effectiveness of multilevel supervision in enhancing feature representation learning. The MIM module further enables change moment identification while simultaneously improving monitoring performance by establishing a direct connection between continuous change monitoring and change moment inference. When SSL and MIM are jointly employed, the model achieves substantial gains in both tasks. Incorporating all three components yields the best overall performance, highlighting their complementary roles. Similar trends are observed on SpaceNet7.

## IV. DISCUSSION

### A. Discussion About the Run Time

CCM-Net merges the temporal and batch dimensions into a unified batch for spatial feature extraction, improving computational efficiency. We compare this strategy with a conventional Siamese implementation on DynamicEarthNet and SpaceNet7 (Table V). The Siamese model processes each time step sequentially using shared parameters, whereas the merged strategy processes all time steps simultaneously with the same extractor. Although both approaches are mathematically equivalent in spatial feature extraction, the merged implementation better leverages GPU parallelism and avoids iterative overhead, resulting in faster training and inference.

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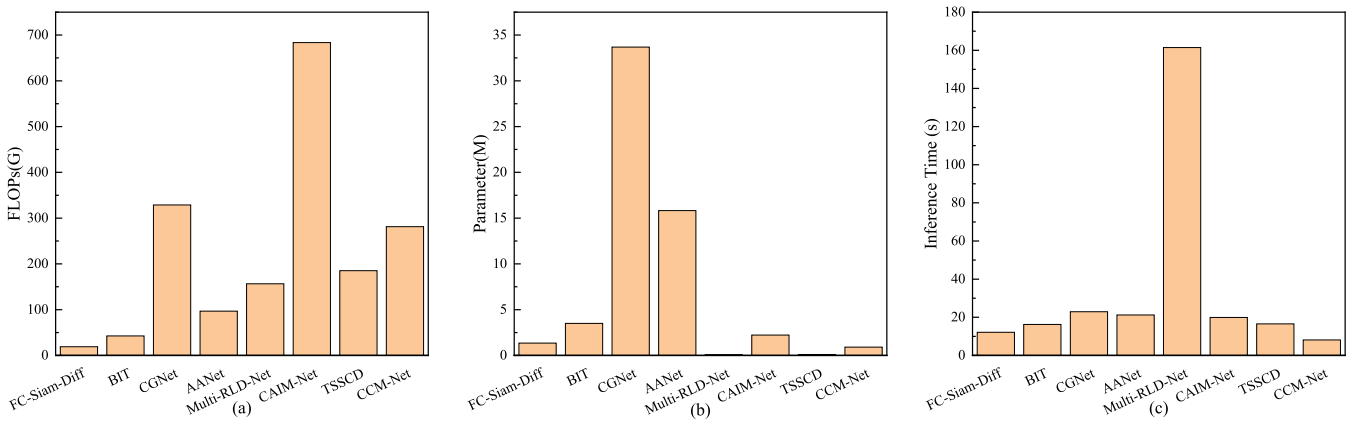



Fig. 12. FLOPs, parameters and inference time of different methods on SpaceNet7 dataset. (a) FLOPs. (b) Parameter. (c) Inference time.

TABLE V
TRAIN AND INFERENCE TIME OF THE CCM-NET (SIAMESE) AND PROPOSED CCM-NET ON BOTH DATASETS

| | DynamicEarthNet | | SpaceNet7 | |
|---|---|---|---|---|
| | Training | Inference | Training | Inference |
| CCM-Net(Siamese) | 346.35s | 12.03s | 1247.75s | 33.40s |
| CCM-Net | 171.51s | 8.15s | 655.43s | 21.60s |

### B. Discussion About the Robustness

Clouds, shadows, temporal misalignment, and seasonal variations may introduce false changes in TSIs. Although these factors are not explicitly modeled in CCM-Net, the proposed SSL module captures long-range spatiotemporal correlations across multiple observations, while the MIM module infers change moments from continuous monitoring results over the entire sequence. These designs help reduce the influence of transient disturbances and improve temporal consistency. Nevertheless, severe cloud contamination and large temporal misalignment may still affect the detection performance.

### C. Discussion About the Efficiency

FLOPs, parameter count, and inference time are important indicators of computational efficiency. Fig. 12 compares the computational cost of representative deep learning methods and the proposed CCM-Net on SpaceNet7. Although CCM-Net exhibits higher computational complexity than bi-temporal methods such as FC-Siam-Diff and BIT, it remains more efficient than most TSCD approaches and is lighter than CG-Net. In terms of model size, CCM-Net is comparable to FC-Siam-Diff and only slightly larger than Multi-RLD-Net and TSSCD, indicating that the proposed SSL and MIM modules introduce only a modest increase in parameters. Regarding inference efficiency, bi-temporal methods require repeated processing of adjacent image pairs, while patch-based strategies further increase latency. In contrast, CCM-Net directly models the entire temporal sequence in a single forward pass, avoiding redundant computations and achieving the shortest inference time among the compared methods. Overall, CCM-Net achieves a favorable tradeoff between detection accuracy and computational efficiency, demonstrating its ability to capture long-term temporal dependencies without imposing excessive computational burden. These results suggest that CCM-Net is well suited for large-scale remote sensing change monitoring applications.

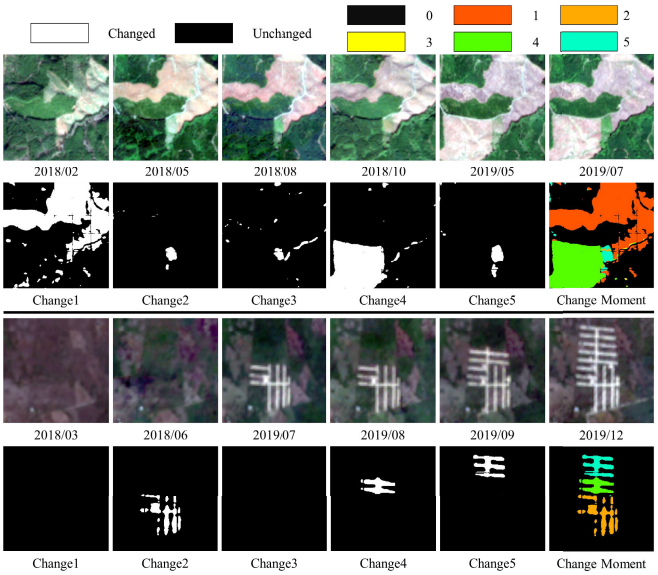



Fig. 13. Visualization results for verifying the model's generalization ability.

### D. Discussion About the Model Generalization

To evaluate the generalization of the proposed CCM-Net, we conducted experiments on the DynamicEarthNet dataset, which includes 20 previously unseen and unlabeled TSIs blocks, providing an important opportunity to assess the model's performance in real-world, unseen scenarios. As shown in Fig. 13, CCM-Net accurately monitors continuous changes and identifies change moments even in regions not seen during training, demonstrating the robustness and applicability of the framework.

### E. Discussion About Failure Cases and Limitations

Although CCM-Net achieves competitive performance on both DynamicEarthNet and SpaceNet7, several failure cases

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can still be observed in Figs. 10 and 11. In particular, the examples shown in the last column reveal noticeable omission errors, where portions of change regions remain undetected. These missed detections mainly occur in thin boundary structures and isolated change patches. Since such changes occupy only a few pixels and often exhibit subtle differences from surrounding areas, they tend to generate weak spatiotemporal responses. As a result, some change information may be suppressed during feature aggregation. In addition, the pronounced class imbalance in SpaceNet7 further limits the learning of discriminative representations for rare change samples, increasing the likelihood of missed detections. These factors may lead to incomplete change monitoring results and inaccurate change moment identification in certain cases. Future work will explore fine-grained feature learning and imbalance-aware optimization strategies to improve the detection of small and sparse changes.

## V. Conclusion

In this study, we propose CCM-Net, a novel deep learning approach for continuous change monitoring in TSIs. CCM-Net leverages the difference features between temporally adjacent images to monitor continuous changes and identify change moments. To the best of our knowledge, this is the first deep learning-based method for directly monitoring continuous changes in TSIs.

Our experiments use TSIs as input data for the network. The CCM-Net model is a fully convolutional network, guided by densely annotated change information, enabling it to learn an end-to-end mapping that transforms spectral differences in time series into continuous change monitoring and change moment identification. We evaluated CCM-Net's accuracy in two global-scale study areas, achieving superior results compared to traditional change monitoring methods. Additionally, we discuss the model's structure, designed to capture spatial and spatiotemporal correlations, and explore a suitable loss function for continuous change monitoring and change moment identification. This joint detection approach bridges a significant gap in TSCD by enabling continuous change monitoring with deep learning, addressing both the "when" and "where" of changes.

Despite its advantages, the CCM-Net framework has limitations that highlight future research directions. A key challenge is its reliance on large amounts of annotated data for training. While it performs well under supervised settings, future work will explore extending CCM-Net to a semi-supervised framework, reducing dependence on labeled data and enhancing its applicability in low-resource environments. Furthermore, from a practical perspective, it is crucial not only to monitor continuous changes but also to identify and classify the types of these changes, addressing the challenges of "when," "where," and "what" changes have occurred.

## Acknowledgment

The numerical calculations in this article have been done on the supercomputing system in the Supercomputing Center of Wuhan University.

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